CHANDU PRASAD BHAIRAPU



AI Developer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in full-stack features with Python, FastAPI and React, APIs and integrations, and AI-native development with coding agents. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by experience building and deploying applications on Azure and AWS in start-up and enterprise teams.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Developed Expert Interface end to end, a graph curation web app (React frontend, Python microservices backend with APIs) deployed on Azure, turning the needs of SAP domain experts into features for validating and approving knowledge-graph changes.
- Built software as an AI-augmented engineer with AI coding agents (Claude Code), custom skills and agentic coding workflows as a standard part of professional development, doing code review of AI-generated code for correctness and applying clean code principles.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph for millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Sole developer owning the full knowledge-extraction pipeline end-to-end, from requirements through data ingestion, chunking, prompt engineering and indexing (agentic, deterministic and monolithic strategies), reporting directly to the CTO in a small team.
- Designed and optimized multi-agent orchestration with LangGraph (entity-extraction, relationship-analysis, consolidation and reflection agents running in parallel) for complex regulatory and technical documents, raising precision/recall from 33%/45% to 93%/90% against an SAP-system-derived ground truth.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- **Tech stack: Python, FastAPI, Python microservices, React, Postgres, Docker, Azure, Claude Code, MCP, LangGraph, LangChain, LiteLLM, RAG, GraphRAG, Neo4j, MongoDB, Kubernetes**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Developed a React front-end and a Python backend with FastAPI for an agentic AI application, translating the requirements of business stakeholders into the solution, and tested the APIs with Postman.
- Designed and deployed an AI agent with LangGraph that routes each request across vector database queries, general LLM inference and summarization.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK, following clean code practices; version-controlled in GitLab with CI/CD.
- **Tech stack: Python, FastAPI, React, Postman, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD, LangGraph, LangChain**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 | Leverkusen, Germany

- Designed Retrieval-Augmented Generation (RAG) pipelines, focusing on chunking strategies, vector databases, indexing, and fine-tuning models for optimized retrieval.
- Automated RAG generation and retrieval evaluation with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- Built AWS cloud infrastructure with Lambda, SQS and DynamoDB for PrismaChat RAG evaluation using CDK.
- **Tech stack: Python, LangChain, Hugging Face Transformers, PyTorch, RAGAS, PytrecEval, Vector Databases, Azure AI Search, AWS (Lambda, SQS, DynamoDB, CDK), Docker, Streamlit**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) to feed the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, SciPy, Matplotlib, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XG-Boost regressor, FB Prophet, LSTM models for time series analysis for EVs.
- Developed LSTM, achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Client Self-Serve Portal | TypeScript | NextJS | NestJS | FastAPI | PostgreSQL

- Self-serve portal where advertiser clients create campaigns, track delivery and download reports, with a NextJS front end, a NestJS API with typed DTOs and an OpenAPI spec, and a FastAPI service for pricing and forecasts, over seeded synthetic campaign data in PostgreSQL.

TECHNICAL SKILLSET

- **Backend & APIs:** Python, FastAPI, Python microservices, API design, API testing (Postman), React (beginner)
- **Data:** Postgres, SQL, MongoDB, Neo4j, Vector Databases (Qdrant, Pinecone, Azure AI Search)
- **Cloud & DevOps:** Docker, AWS (Lambda, SQS, DynamoDB, EC2, CDK), Azure, Kubernetes, Git, GitLab CI/CD
- **AI-native & LLM:** Claude Code (coding agents, custom skills), MCP, RAG, GraphRAG, LangGraph, LangChain, LiteLLM
- **Languages:** English (Business Fluent), German (B1), Hindi (Fluent), Telugu (Native)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering (Minor in Computer Science & Engineering) — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India